

ODD 2024
Tutorial Sheet – 4 & 5
Software Development Fundamentals – I (15B11CI111)

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration, and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real-world problems	Apply (Level 3)

Note: Students are advised to submit their solutions to their respective tutorial faculties

Q1. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    int i;
    for(i = 0; i>9; i+=3)
    {
        printf("for ");
    }
    return 0;
}
```

Q2. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    int i = 1, j = 1;
    for(--i && j++ ; i<10; i+=2)
    {
        printf("loop ");
    }
    return 0;
}
```

Q3. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    for(5;2;2)
        printf("Hello");
    return 0;
}
```

Q4. What will be the output of the C program?

```
#include<stdio.h>
int main()
```

```

{
    int n;
    for (n = 9; n!=0; n--)
        printf("n = %d", n--);
    return 0;
}

```

Q5. What will be the output of the C program?

```

#include <stdio.h>
int main()
{
    int c = 5, no = 10;
    do {
        no /= c;
    } while(c--);

    printf ("%dn", no);
    return 0;
}

```

Q6: Write C programs to print the following patterns:

(a) <pre> * </pre>	(b) <pre> 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1 </pre>
(c) <pre> A B C D E F G H I A B C D E F G A B C D E A B C A </pre>	(d) <pre> * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * </pre>

Q7. Write C Programs to do the following:

- (i) Write a program in C language for printing all the odd / even numbers within a given range using the while loop.
- (ii) Write a program in C language to print the addition of numbers within a range.
- (iii) Write a program in C language to print the factorial of a number.
- (iv) Write a program in C language to print all the uppercase letters.
- (v) Write a program in C language to print the multiplication table of an integer.
- (vi) Write a program to print the square roots of all the numbers within a range.
- (vii) Write a program in C to print a range of numbers (between **a** to **b**) that are divisible by 5.
- (viii) Write a program in C to print arithmetic progression and its sum.
- (ix) Write a C program to print all Armstrong Numbers from 1 to N.